CSE 4304-Data Structures Lab. Winter 2022
Date: October 22, 2022.
Target Group: All
Topic: Hashing

Instructions:

- Task naming format: fullID_T01L10_2A.c/CPP
- If you find any issues in the problem description/test cases, comment in the google classroom.
- If you find any tricky test case that I didn't include but others might forget to handle, please comment! I'll be happy to add.
- Use appropriate comments in your code. This will help you to recall the solution in the future easily.
- Obtained marks will vary based on the efficiency of the solution.
- Do not use <bits/stdc++.h> library, add the necessary libraries one after another.
- Do not use any built-in functions or any STL libraries. If the solution is extremely difficult without the usage of such components, discuss it with the instructors.
- Modified sections will be marked with BLUE color.

Group	**Tasks**
1B	1 2 3 4
2B	1 2 3 4
2A	1 2 3 4
1A	1 2 3 4

Task 1

Given two list of integers, your task is to find whether list-2 is a subset of list-1 or not. Both lists are **unsorted** and consist of **distinct** elements.

There will be two lines of input representing the lists. Both lines end with -1 representing the end of that list.

Output:

- 'Yes': if list-2 is a subset of list-1
- 'No': if list-2 is not a subset of list-1

Sample Input	**Sample Output**
12 14 9 7 22 1 6 0 45 39 5 -1 7 5 6 -1	Yes
7 5 6 -1 12 14 9 7 22 1 6 0 45 39 5 -1	No
1 2 9 88 41 0 17 11 6 -1 100 52 2 8 -1	No

Constraint: Provide O(n) solution.

Note:

- Explore the 'unordered map' library function (https://www.geeksforgeeks.org/unordered_map-in-cpp-stl/)
- **Rejected solution**: Store the two lists in two arrays. Write an O(n^2) loop to check each element of list-2 with respect to every element of list-1.
- Provide O(n) solution with Hashmaps

Task 2

Given a collection of integers and a number ‘target’, find the pairs of integers whose summation is equal to ‘target’. The elements of the collection may not be unique.

The first line provides the collection of integers where -1 denotes the end of the input. The following line will contain the target value.

Output:

- Print the pairs whose summation equals ‘target’.
- If none of the pairs adds up to ‘target’, print ‘No pairs found’.

Sample Input	**Sample Output**
2 5 4 12 9 1 3 17 11 8 -1 13	(8,5), (12,1), (9,4), (11,2)
2 5 4 2 0 1 3 -1 4	(2,2), (3,1), (4,0)
2 5 4 2 0 2 7 -1 6	(4,2)
2 5 4 12 9 1 3 17 8 11 8 5 -1 13	(8,5), (12,1), (9,4), (11,2), (5,8)
1 1 1 2 2 2 3	(1,2), (1,2), (1,2)
1 1 1 1 2 2 1 1 1 1 2 3	(1,2), (1,2), (1,2)
4 -2 2 7 9 1 3 1 0 -1 7	(4,3), (7,0), (9,-2)
2 5 4 12 9 1 3 17 11 8 10 -1 100	No pairs found

Note:

- The elements/pairs can appear in any order. Hence [(8,5), (12,1), (9,4), (11,2)] and [(5,8), (1,12), (4,9), (2,11)] means the same. There can be many other combinations.
- **Rejected Solution:** Store the collection in an array. For every element, search the (target-current) element from the remaining portion of the array. Complexity O(n^2).
- Provide O(n) solution with Hashmaps

Task 3:

Given a sentence, you have to find the word(s) that occur more than once. Ignore that punctuation marks.

Input	**Output**
data atad structure atad data	data 2 atad 2
I know you know this, but you do not know of unknown trolls, because no known trolls will sew by windows, though they will owe you a hello when they throw a hoe, as it will go low and blow a hole in that window and so it will follow, that it happened awhile ago, as a troll will stand on a knoll and show you how to throw snow tomorrow at a rhino named Joe, who plays piano as he sips a cappucino and sings soprano in an inferno caused by a volcano in Reno with a casino at the bottom. Of the volcano.	volcano 2 a 9 by 2 will 5 at 2 throw 2 in 3 you 4 it 3 the 2 know 3 and 4 trolls 2 they 2 as 3 that 2
I know you know this but you do not know of unknown trolls because no known trolls will sew by windows though they will owe you a hello when they throw a hoe as it will go low and blow a hole in that window and so it will follow that it happened awhile ago as a troll will stand on a knoll and show you how to throw snow tomorrow at a rhino named Joe who plays piano as he sips a cappucino and sings soprano in an inferno caused by a volcano in Reno with a casino at the bottom. Of the volcano	volcano 2 a 9 by 2 will 5 at 2 throw 2 in 3 you 4 it 3 the 2 know 3 and 4 trolls 2 they 2 as 3 that 2
This refers to an exam where James had written ‘had had’ where John had written just ‘ had’. The examiner had approved James’ version.	James 2 where 2 had 6 written 2
This refers to an exam where James had written had had where John had written just had The examiner had approved James version	James 2 where 2 had 6 written 2

Note:

- Show the words in any order.
- May use **getline**(cin, sentence) to take the input.

Task 4:
Given a sentence and a word, your task is to print the words that consist of the same **unique character set.**

The first line of input will contain the sentence, and the second line will contain the word. Print the word(s) that consist of the same unique character set (length doesn't matter)

Input	**Output**
You may know the answer but it is not yam or maaayaaay or yammy may	may yam maayaaay yammy
student will act like students and it is studentish dont write studnet by mistake student	student students studnet
abcd abd acd bad baad baacd aabbccdd accc aadd abbe aaag aabbcd	abcd baacd aabbccdd
aceg aceegg ggccaaaee bcdg hbae aceg	aceg aceegg ggccaaaee

Note:

- Write a hash function which will check the similarity between two strings utilizing the unique characters. Just adding the ASCII of the unique characters won't work as 'ad' and 'bc' gives the same result. Need a better hash value.
- Explanation for input-1: The unique character set in 'may' is 'm,a,y'. As 'maaayaaay' also consists of the same set of unique characters 'm-a-y' it is a part of the output.
- All the unique characters of the word must be used. Hence, the word 'aabbcd'- 'abd', 'acd' etc is not a match as one of the unique characters are missing.
- The hash value of two words having the same unique character set should be equal.
- May use **getline**(cin, sentence) to take the input.